a) Step 1: Calculate available memory

Total RAM = 96 GB

Reserved RAM = 8 GB

available = 96 − 8 = 88 GB

Total allocated vCPU

Total vCPU = 12 × 3 = 36 vCPUs

CPU cores = 24

ratio = $\frac{36}{24}$ = 1.5 : 1

Calculate memory over-commitment ratio

Total configured VM memory:

= 12 × 8 = 96 GB

available = 88

memory ratio = $\frac{96}{88}$ = 1.09 : 1

b) average memory sufficiency

total memory = 12 × 5 = 60 GB

available = 88 GB

Free memory: 88 − 60 = 28 GB

Peak memory sufficiency

Peak memory = (8 × 8) + (4 × 6)

= 64 + 24

= 88 GB

Free memory,

88 − 88 = 0 GB

Average CPU sufficiency

Each VM - 3 CPU

Total allocated vCPU = 36

Normal utilization = 35%

CPU usage = $36 \times 0.35$

$= 12.6$ cores

Physical CPU cores = 24

Available CPU capacity

$= 24 - 12.6$

$= 11.4$ cores

Average CPU utilization:

$= \frac{12.6}{24} \times 100$

$= 52.5\%$

d) Peak CPU sufficiency

Peak CPU utilization = 75%.

Peak CPU = $36 \times 0.75 = 27$ cores

Physical CPU cores = 24

CPU shortage: $27 - 24 = 3$ cores

Peak CPU utilization:

$= \frac{27}{24} \times 100$

$= 112.5\%$

CPU contention - Add physical CPU or other host

Memory exhaustion - Increase RAM and monitor memory usage

VM performance degradation - Use resource reservation and limits
Data leakage - Apply encryption
Single host failure - Implement high availability and backup

4. CLOUD FEDERATION, SDDC and SOA

Cloud Federation is integration of multiple independent cloud environment to share resources

It wants to integrate:
* Learning Management system
* Digital library
* Student information system.

Role of cloud Federation:

Working:

1. Each location operates its own cloud data centre
2. The cloud platforms are connected through secure networks
3. Resources and applications are shared
4. Students can access services from diff locations
5. Workloads can be distributed across data centres.

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Section 1 - learning management system
Section 2 - Digital library
Section 3 - Student information system

Role of SDDC:

* SDDC + Software Defined Data network

1. Software - Defined compute - Allocates CPU and memory
2. software - Defined storage - manages student documents and learning materials
3. Software - Defined Networking - connects diff cloud platforms
4. Automation - Automatically manages resources
5. centralized Management - Monitors all data centres

Advantages:

* Dynamic resource allocation
* Easy management
* Improved Scalability
* Better resource utilization
* Automatic Workload balancing

Example: During online examinations, additional computing resources can be allocated to the LMS

Role of SOA:

SOA - Service - Oriented Architecture

5. Working:
1. Each application provides its services through APIs
2. Other applications send requests to those services
3. The service processes the request
4. The result is returned to the requesting applications

Ex:
* LMS provides course details
* SIS provides student information

Challenges and solutions:

A. Federated Identity:
Diff cloud platforms may use diff login systems
problem: Students may need multiple usernames and passwords.
solution:
* Implement single sign-on
* Use OAuth 2.0 and OpenID Connect

B. Users presence information
Problem: Sharing this info across cloud platforms may Expose private details
solution:
* Shares only necessary presence information
* Obtain user consent

SAVEETHA

...ration:

Problem: Data protection laws and institutional policies may differ

Solution:

* Use approved data storage locations
* Maintain data residency
* Encrypt data during storage and transmission

D. Privacy And security

Problem: Unauthorized users may access student records and academic information

Solution:

* Multi-factor authentication
* Role-Based Access control
* Security monitoring

Additional solutions for Reliable Integration

1. Use backup data centres
2. Implement disaster recovery
3. Use load balancing
4. Use standard APIs
5. Monitor network performance
6. Maintain regular database backups

5. Given
Cameras = 150
Clip size = 11 MB
One clip every 3 mins
Storage = ₹1.50/GB-month
Servernet charge = 2000015
Edge filtering = 25%
1GB = 1000MB

a) Daily Data Without Filtering

clips per camera:
(60/3) x 24 = 480

TOTAL clips:
150 x 480 = 72000

Daily data:
72000 x 11 = 288000MB
= 288 GB

Bandwidth:
288000/86400 = 3.33 MB/s.

b) Monthly storage cost
288 x 30 = 8640 GB
8640 x 1.50 = ₹12,960

After edge filtering
uploaded clips:
72000 x 25% = 18000

Daily data:
18000 x 11 = 72 GB

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SAVEETHA

Bandwidth:
4200/86400 × 8 = 0.67 Mbps

Monthly Storage:
72 × 30 = 2160 GB

Storage cost:
2160 × 1.50 = ₹3,240

Monthly saving:
12960 − 3240 = ₹9,720

d) Serverless invocation cost

Monthly invocations:
18000 × 30 = 540000

cost:
540000 × 0.00015 = ₹81

REST API:
Allows cameras to upload vedio clips to cloud storage using HTTP requests.

Object Storage:
Suitable because it stores large unstructured files, is scalable and cost-effective

conclusion:
Edge filtering reduces data transfer, storage cost and serverless charges